The z-Transform

Signals and Linear Systems

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Outline

- **Introduction:** Why the z-transform is the discrete-time Laplace transform
- **The z-Transform Definition**
- **The Region of Convergence (ROC):**
  - Right-sided, left-sided, and two-sided sequences
  - Poles, zeros, and the unit circle
  - ROC properties and significance
- **The Inverse z-Transform:**
  - Formal contour formula
  - Partial fractions and power-series inversion
- **Properties of the z-Transform**
- **LTI Systems, Causality, Stability, and Difference Equations**

Introduction and Definition

Motivation: From Fourier to a Broader Transform

Continuous Time

Fourier: $e^{j\omega t}$ is an eigenfunction of LTI systems

$$x(t) = e^{j\omega t} \implies y(t) = H(j\omega)\, e^{j\omega t}$$

Laplace: generalize to $s = \sigma + j\omega$

$$x(t) = e^{st} \implies y(t) = H(s)\, e^{st}$$

Discrete Time

DTFT: $e^{j\omega n}$ is an eigenfunction of LTI systems

$$x[n] = e^{j\omega n} \implies y[n] = H(e^{j\omega})\, e^{j\omega n}$$

z-Transform: generalize to $z = re^{j\omega}$

$$x[n] = z^n \implies y[n] = H(z)\, z^n$$

The Pattern

In both domains, Fourier analysis restricts us to the frequency axis. The broader transform (Laplace / z) allows a **complex** variable, letting us analyze unstable systems and characterize causality and stability.

The Eigenfunction Property of z^n

- For a discrete-time LTI system with impulse response $h[n]$, setting $x[n] = z^n$, the eigenvalue is the z-transform of $h[n]$:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k]\, z^{-k} = H(z)\, z^n.$$

The z-Transform

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

The z-transform is to the **DTFT** what the Laplace transform is to the **CTFT**.

What It Gives Us

- Extends analysis to a broader class of signals
- Lets us analyze **unstable** systems
- Reveals causality & stability via poles, zeros, and ROC

Geometric Difference

- s-plane: Fourier transform lives on the $j\omega$**-axis**
- z-plane: DTFT lives on the **unit circle**

Sampling e^{st} every T seconds gives

$$e^{s\,nT} = \left(e^{sT}\right)^n$$

Define

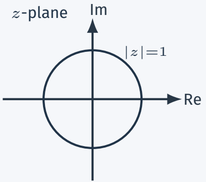
$$z \triangleq e^{sT}$$

So the sampled signal is z^n.

The Change of Variable

$$z \;\triangleq\; e^{sT} \;=\; e^{\sigma T}\,e^{j\omega T} \;=\; r\,e^{j\theta}$$

$$|z| = r = e^{\sigma T} \leftrightarrow \text{damping}$$
$$\angle z = \theta = \omega T \leftrightarrow \text{frequency}$$



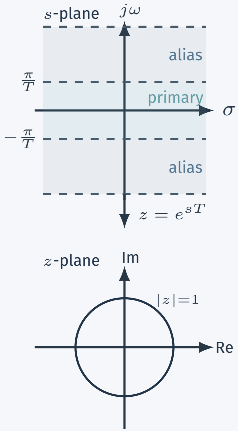
Why z instead of e^s?

- In DTFT, we set $T = 1$ and work with $e^{j\omega n}$. We could generalize the same way and write e^{sn}, $s = \sigma + j\omega$.
- **But the map** $s \mapsto e^{sT} = e^{\sigma T}\, e^{j\omega T}$ **is many-to-one:** shifting ω by $\frac{2\pi}{T}$ gives the same $z = e^{sT}$.
- In continuous time, $e^{j(\omega+2\pi/T)t} \neq e^{j\omega t}$ for general t, so distinct s values still give distinct signals.
- In discrete time, n is an **integer**, so $e^{j(\omega+2\pi/T)Tn} = e^{j\omega Tn}$—different s values produce the same discrete-time signal.
- Using z as a parameter solves this redundancy.

z Parameterization

Parameterizing by z collapses the redundancy in the s-plane. Also, as we will see, using z makes the algebra and geometry of the transform much more intuitive.

Many-to-one mapping



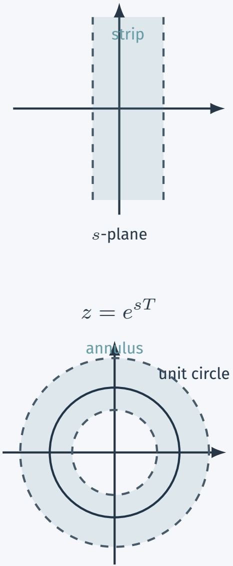
How the Mapping $z = e^{sT}$ Changes the Geometry

The mapping $z = e^{sT}$ transforms s-plane geometry into z-plane geometry:

s**-plane**	$\longrightarrow$	z**-plane**
Vertical line $\text{Re}\{s\} = \sigma_0$	$\longrightarrow$	Circle $\lvert z\rvert = e^{\sigma_0 T}$
$j\omega$-axis ($\sigma_0 = 0$)	$\longrightarrow$	Unit circle $\lvert z\rvert = 1$
ROC strip	$\longrightarrow$	ROC annulus

Consequence

The unit circle plays the same role in the z-plane that the $j\omega$-axis plays in the s-plane. ROC questions become questions about **radii**, not real parts.



The System Function $H(z)$

Since z^n is an eigenfunction of DT-LTI systems, the system is completely characterized by its **system function**:

System Function

$$H(z) = \sum_{k=-\infty}^{\infty} h[k]\, z^{-k} \qquad x[n] = z^n \implies y[n] = H(z)\, z^n$$

What $H(z)$ Encodes

- Convolution in time $\longrightarrow$ multiplication in z
$$Y(z) = H(z)\, X(z)$$
- Poles of $H(z)$ $\longrightarrow$ natural modes of the system
- Zeros of $H(z)$ $\longrightarrow$ frequencies the system blocks

Parallel with Laplace

Laplace	z-transform
$H(s)$	$H(z)$
$Y(s) = H(s)X(s)$	$Y(z) = H(z)X(z)$

Definition of the z-Transform and Its DTFT Connection

z-Transform

$$X(z) \triangleq \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Write

$$z = re^{j\omega}.$$

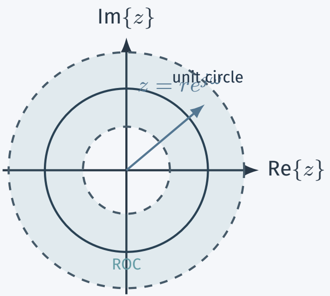
Then

$$\begin{aligned} X(re^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left(x[n]r^{-n}\right)e^{-j\omega n}. \end{aligned}$$

Hence

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}.$$

Setting $r = 1$ restricts the z-transform to the unit circle, so we recover the DTFT. Therefore the DTFT exists iff the ROC includes $|z| = 1$.



Fourier and z

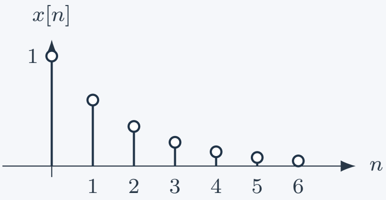
The DTFT is the restriction of the z-transform to $|z| = 1$.

Examples and the ROC

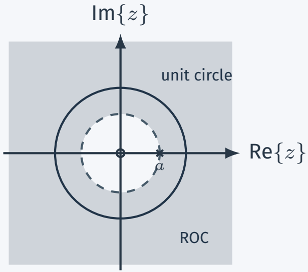
Consider
$$x[n] = a^n u[n].$$
Then
$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n, \qquad |az^{-1}| < 1$$
$$= \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \qquad |z| > |a|.$$

- This is the discrete-time analogue of $e^{-at}u(t)$ in the Laplace chapter.
- Pole at $z = a$, zero at $z = 0$.
- The ROC is the exterior of a circle.
- If $|a| < 1$, the ROC includes the unit circle, so the DTFT exists.

Sequence for $a = 0.6$



Pole-zero plot and ROC



Example 10.2: A Left-Sided Exponential

Consider

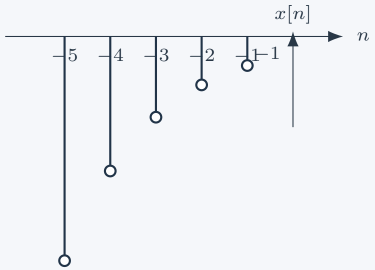
$$x[n] = -a^n u[-n-1].$$

Then

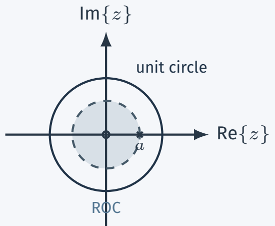
$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{-1} -a^n z^{-n} \\ &= -\sum_{m=1}^{\infty} a^{-m} z^m = -\sum_{m=1}^{\infty} (a^{-1}z)^m \\ &= \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \qquad |z| < |a|. \end{aligned}$$

- Same algebraic expression as Example 10.1.
- But now the ROC is the interior of the circle.
- This is exactly the same message we saw with Laplace transforms: algebraic form alone is not enough.

Sequence for $a = 0.6$



Pole-zero plot and ROC



Why the ROC Matters

- Examples 10.1 and 10.2 have the same

$$X(z) = \frac{z}{z-a},$$

  but they are different sequences because the ROCs differ.

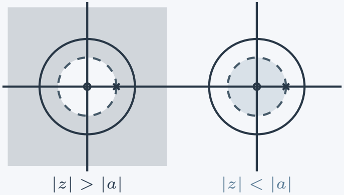
- So a z-transform is specified by

$$\text{algebraic expression} + \text{ROC}.$$

- For rational transforms:
  - **zeros** are the roots of the numerator,
  - **poles** are the roots of the denominator.
- The ROC tells us:
  - whether the sequence is right-sided, left-sided, or two-sided,
  - whether the DTFT exists,
  - and later, whether an LTI system is causal or stable.

Laplace: half-planes or strips $\Longleftrightarrow$ z-transform: disks, exteriors, annuli

Same poles and zeros, different ROC



Example 10.3: Sum of Two Exponentials

$$x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n].$$

By linearity and Example 10.1,

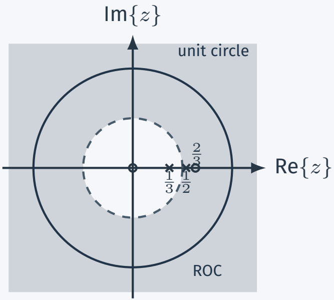
$$X(z) = \frac{7}{1-\frac{1}{3}z^{-1}} - \frac{6}{1-\frac{1}{2}z^{-1}}$$
$$= \frac{1-\frac{3}{2}z^{-1}}{\left(1-\frac{1}{3}z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)}$$
$$= \frac{z\left(z-\frac{2}{3}\right)}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}.$$

Since both components are right-sided,

$$\text{ROC}: |z| > \frac{1}{2}.$$

- Poles at $z = \frac{1}{3}, \frac{1}{2}$.
- Zeros at $z = 0, \frac{2}{3}$.
- The ROC is outside the outermost pole, exactly as in the Laplace chapter.

Pole-zero plot



Consider

$$x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right) u[n].$$

Use Euler's identity:

$$\sin\left(\frac{\pi}{4}n\right) = \frac{e^{j\pi n/4} - e^{-j\pi n/4}}{2j}.$$

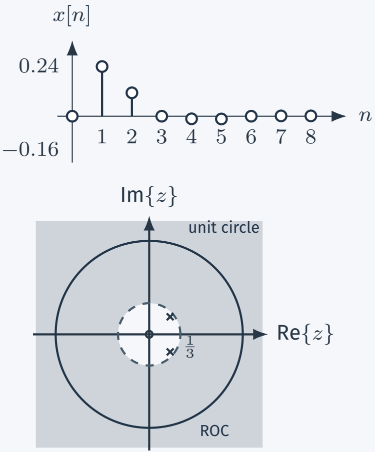
Then

$$\begin{aligned} X(z) &= \frac{1}{2j}\frac{1}{1-\frac{1}{3}e^{j\pi/4}z^{-1}} - \frac{1}{2j}\frac{1}{1-\frac{1}{3}e^{-j\pi/4}z^{-1}} \\ &= \frac{\frac{\sqrt{2}}{3}z^{-1}}{1-\frac{\sqrt{2}}{3}z^{-1}+\frac{1}{9}z^{-2}} \\ &= \frac{\frac{\sqrt{2}}{3}z}{\left(z-\frac{1}{3}e^{j\pi/4}\right)\left(z-\frac{1}{3}e^{-j\pi/4}\right)}. \end{aligned}$$

Both exponential components are right-sided, so

$$\text{ROC}: |z| > \frac{1}{3}.$$

Sequence and ROC



ROC Properties 1–2

Property 1

The ROC of $X(z)$ is a **ring centered at the origin**

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]r^{-n}e^{-j\omega n}, \qquad \sum_{n=-\infty}^{\infty} |x[n]|r^{-n} < \infty.$$

Absolute convergence depends on the radius $r = |z|$, not on the angle.

Property 2

The ROC cannot contain any poles.

At a pole, $X(z)$ is unbounded. Zeros may lie in the ROC; poles may only lie on its boundary or outside it.

ROC Property 3: Finite-Duration Sequences

Property 3

If $x[n]$ is finite-duration, the ROC is the entire z-plane, except possibly at $z = 0$ and/or $z = \infty$.

Justification: Only finitely many terms appear:

$$X(z) = \sum_{n=N_1}^{N_2} x[n]z^{-n}.$$

Thus every **finite, nonzero** z is safe.

But note there are two special cases:

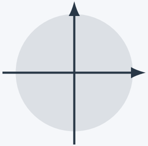
- A sample at $n > 0$ creates negative powers of z and can exclude $z = 0$.
  Since as $|z| \to 0$, $z^{-n} \to \infty$ for $n > 0$.
- A sample at $n < 0$ creates positive powers of z and can exclude $z = \infty$.
  Since as $|z| \to \infty$, $z^{-n} \to \infty$ for $n < 0$.

As a result, if $N_1 \geq 0$, the ROC includes ∞; if $N_2 \leq 0$, it includes 0.

Impulse

$$\delta[n] \stackrel{\mathcal{Z}}{\longleftrightarrow} \sum_{n=-\infty}^{\infty} \delta[n] z^{-n} = 1.$$

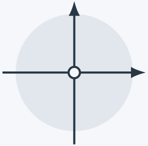
ROC: entire z-plane, including 0 and ∞.



Delayed impulse

$$\delta[n-1] \stackrel{\mathcal{Z}}{\longleftrightarrow} \sum_{n=-\infty}^{\infty} \delta[n-1] z^{-n} = z^{-1}.$$

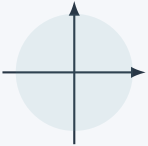
ROC: entire z-plane except $z = 0$.



Advanced impulse

$$\delta[n+1] \stackrel{\mathcal{Z}}{\longleftrightarrow} z$$

ROC: entire finite z-plane, but not $z = \infty$.



These examples show how positive powers of z threaten ∞ and negative powers threaten 0.

ROC Property 4: Right-Sided Sequences

Property 4

If $x[n]$ is a right-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all finite values of z for which $|z| > r_0$ will also be in the ROC.

Right-sided means $x[n] = 0$ for $n < N_1$, so

$$X(z) = \sum_{n=N_1}^{\infty} x[n] z^{-n}, \qquad \sum_{n=N_1}^{\infty} |x[n]| r_0^{-n} < \infty$$

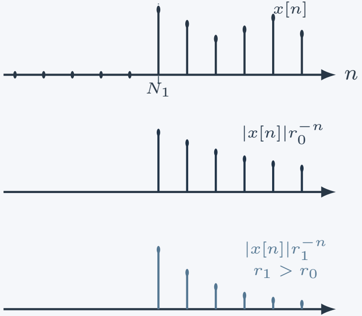
when $|z| = r_0$ lies in the ROC. For any $r_1 > r_0$:

$$|x[n]| r_1^{-n} = |x[n]| r_0^{-n} \left(\frac{r_0}{r_1}\right)^n .$$

Since $0 < r_0/r_1 < 1$, the extra factor decays with $n \Rightarrow$ absolute summability at r_1 is guaranteed.

Endpoint nuance: if $N_1 < 0$, ∞ may be excluded; for causal sequences ($N_1 \geq 0$), $\infty \in$ ROC.

Faster decay for $r_1 > r_0$



Property 5

If $x[n]$ is a left-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all values of z for which $0 < |z| < r_0$ will also be in the ROC.

Left-sided means $x[n] = 0$ for $n > N_2$, so

$$X(z) = \sum_{n=-\infty}^{N_2} x[n]z^{-n}, \qquad \sum_{n=-\infty}^{N_2} |x[n]|r_0^{-n} < \infty$$

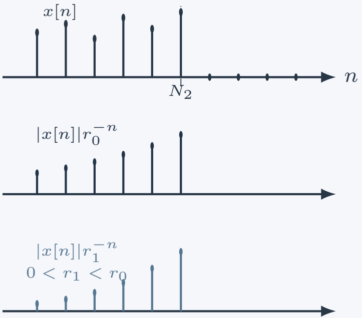
when $|z| = r_0$ lies in the ROC. For any $0 < r_1 < r_0$:

$$|x[n]|r_1^{-n} = |x[n]|r_0^{-n}\left(\frac{r_0}{r_1}\right)^n .$$

Since $r_0/r_1 > 1$ but $n \to -\infty$, the extra factor decays $\Rightarrow$ absolute summability at r_1 is guaranteed.

Endpoint nuance: if $N_2 > 0$, 0 may be excluded; for anticausal sequences ($N_2 \leq 0$), $0 \in$ ROC.

Faster decay for $0 < r_1 < r_0$



ROC Property 6: Two-Sided Sequences

Property 6

If $x[n]$ is two-sided and has a z-transform, the ROC is an annulus.

Split a two-sided sequence into

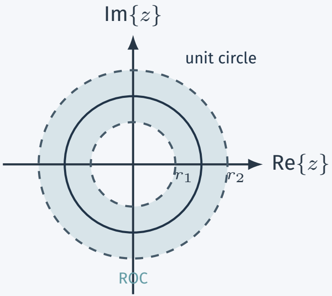
$$x[n] = x[n]u[n-N] + x[n]\big(1 - u[n-N]\big).$$

The right-sided part contributes an exterior ROC, and the left-sided part contributes an interior ROC.

Overlap test

The z-transform exists only if those two ROCs overlap. When they do, the overlap is a ring: $r_1 < |z| < r_2$.

Exterior $\cap$ interior $=$ annulus



Example 10.6: A Finite-Length Exponential

For

$$x[n] = \begin{cases} a^n, & 0 \le n \le N-1,\ a > 0 \\ 0, & \text{otherwise} \end{cases}$$

we have

$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \frac{1-(az^{-1})^N}{1-az^{-1}}$$

$$= \frac{z\left[1-a^N z^{-N}\right]}{z-a}.$$

Since the sequence is finite-duration, the ROC is all z except possibly $z=0$.

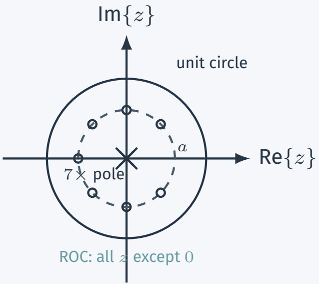
After canceling the factor at $z=a$, the zeros are

$$z_k = a e^{j2\pi k/N}, \qquad k = 1, 2, \ldots, N-1,$$

and there are $N-1$ poles at the origin.

This is a beautiful discrete-time picture: the zeros lie evenly on a circle of radius a.

Pole-zero pattern for $N=8$



Example 10.7: $b^{-|n|}$

Let

$$x[n] = b^{-|n|}, \qquad b > 0.$$

Write it as

$$x[n] = b^{-n}u[n] + b^{n}u[-n-1].$$

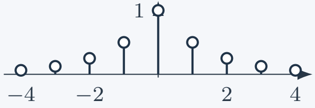
Then the ROC depends strongly on b:

- If $0 < b < 1$, the right-sided part needs $|z| > b$ and the left-sided part needs $|z| < 1/b$. Hence
$$b < |z| < \frac{1}{b}.$$
- If $b > 1$, the required conditions become $|z| > b$ and $|z| < 1/b$, which cannot both hold.
- Therefore **no ROC exists** for $b > 1$.

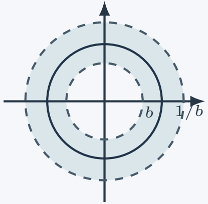
Important lesson

A formal algebraic manipulation can exist, but without a nonempty ROC there is no z-transform.

For $0 < b < 1$



ROC



Property 7

If $X(z)$ is rational, the ROC is bounded by poles or extends all the way to 0 and/or ∞.

Property 8

If $X(z)$ is rational and $x[n]$ is right-sided, the ROC is outside the outermost pole.

More precisely: the ROC is outside the circle through the finite pole of largest magnitude. If the sequence is causal, the ROC also includes $z = \infty$.

Property 9

If $X(z)$ is rational and $x[n]$ is left-sided, the ROC is inside the innermost **nonzero** pole.

More precisely: it extends inward and may include $z = 0$. If the sequence is anticausal, the ROC includes $z = 0$.

Same Algebra / Different Signals

Poles draw the possible circular boundaries. The ROC chooses the side: exterior $\Rightarrow$ right-sided, interior $\Rightarrow$ left-sided, annulus $\Rightarrow$ two-sided.

Example 10.8: Three Possible ROCs

Consider

$$X(z) = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - 2z^{-1}\right)}.$$

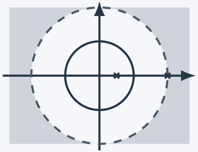
The poles are at

$$z = \frac{1}{2}, \qquad z = 2.$$

Hence there are three possible ROCs:

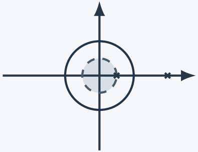
$$|z| > 2, \qquad \frac{1}{2} < |z| < 2, \qquad |z| < \frac{1}{2}.$$

- Exterior ROC: right-sided sequence.
- Interior ROC: left-sided sequence.
- Middle annulus: two-sided sequence.
- Only the middle annulus includes the unit circle, so only that one has a DTFT.



$|z| > 2$

$\frac{1}{2} < |z| < 2$



$|z| < \frac{1}{2}$

The Inverse z-Transform

The Inverse z-Transform

Recall

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}.$$

If the circle $|z| = r$ lies in the ROC, apply the inverse DTFT:

$$x[n]r^{-n} = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) e^{j\omega n} d\omega.$$

Multiply by r^n:

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega})(re^{j\omega})^n d\omega.$$

With $z = re^{j\omega}$ and $dz = jre^{j\omega} d\omega = jz\, d\omega$, we obtain

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1}\, dz$$

where C is any counterclockwise closed contour lying entirely within the ROC.

Inverse z-Transform

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1}\, dz$$

Example 10.9: Partial Fractions with an Exterior ROC

Consider

$$X(z) = \frac{3 - \frac{5}{2}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)}, \qquad |z| > \frac{1}{3}.$$

Partial fractions give

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}.$$

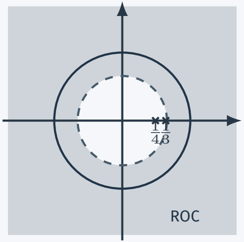
Because the ROC is outside the outermost pole, both terms are right-sided:

$$\frac{1}{1 - \frac{1}{4}z^{-1}} \overset{\mathcal{Z}}{\longleftrightarrow} \left(\frac{1}{4}\right)^n u[n],$$

$$\frac{2}{1 - \frac{1}{3}z^{-1}} \overset{\mathcal{Z}}{\longleftrightarrow} 2\left(\frac{1}{3}\right)^n u[n].$$

Hence

$$x[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n].$$



Main idea

The partial fractions do not change.
The **ROC decides how each term is interpreted**.

Examples 10.10 and 10.11: The Same Algebraic $X(z)$ Gives Different Sequences

Keep the same partial-fraction expansion as Example 10.9:

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}.$$

Example 10.10

ROC:

$$\frac{1}{4} < |z| < \frac{1}{3}.$$

Then the first term is right-sided but the second is left-sided:

$$x[n] = \left(\frac{1}{4}\right)^n u[n] - 2\left(\frac{1}{3}\right)^n u[-n-1].$$

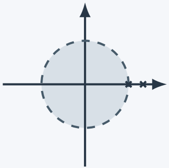
Example 10.11

ROC:

$$|z| < \frac{1}{4}.$$

Now both terms are left-sided:

$$x[n] = -\left(\frac{1}{4}\right)^n u[-n-1] - 2\left(\frac{1}{3}\right)^n u[-n-1].$$



Power-Series Inversion: Examples 10.12 and 10.13

Intuition

Rewrite $X(z)$ as a power series that is valid in the ROC, then match it with

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}.$$

The coefficient of z^{-n} is $x[n]$; positive powers of z therefore represent negative-time samples.

- **Example 10.12**

$$X(z) = 4z^2 + 2 + 3z^{-1} \quad \Longrightarrow \quad x[n] = 4\delta[n+2] + 2\delta[n] + 3\delta[n-1].$$

- **Example 10.13**

$$X(z) = \frac{1}{1 - az^{-1}}.$$

If $|z| > |a|$ ($|az^{-1}| < 1$),

$$X(z) = 1 + az^{-1} + a^2 z^{-2} + \cdots \quad \Longrightarrow \quad x[n] = a^n u[n].$$

If $|z| < |a|$ ($|z/a| < 1$),

$$X(z) = -\frac{z/a}{1 - z/a} = -a^{-1}z - a^{-2}z^2 - a^{-3}z^3 - \cdots \quad \Longrightarrow \quad x[n] = -a^n u[-n-1].$$

Power-Series Inversion: Example 10.14

Consider

$$X(z) = \log(1 + az^{-1}), \qquad |z| > |a|.$$

Using the Taylor series

$$\log(1 + v) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} v^n}{n}, \qquad |v| < 1,$$

with $v = az^{-1}$, we obtain

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n z^{-n}}{n}.$$

Matching powers of z^{-n} gives

$$x[n] = \begin{cases} (-1)^{n+1} \dfrac{a^n}{n}, & n \geq 1 \\ 0, & n \leq 0. \end{cases}$$

Why power series help

The coefficient of z^{-n} in the expansion is exactly the sequence value $x[n]$.

Properties of the z-Transform

Linearity and Time Shifting

Linearity

If

$$x_1[n] \xleftrightarrow{\mathcal{Z}} X_1(z), \qquad x_2[n] \xleftrightarrow{\mathcal{Z}} X_2(z),$$

then

$$ax_1[n] + bx_2[n] \xleftrightarrow{\mathcal{Z}} aX_1(z) + bX_2(z)$$

with ROC containing at least $R_1 \cap R_2$.

Justification: the defining sum is linear. The ROC is at least the intersection; it can be larger if pole-zero cancellation occurs.

Example

$$7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$$

is Example 10.3.

Time Shifting

If

$$x[n] \xleftrightarrow{\mathcal{Z}} X(z),$$

then

$$x[n-n_0] \xleftrightarrow{\mathcal{Z}} z^{-n_0}X(z)$$

with the same ROC, except possibly for adding or deleting 0 or ∞.

Proof sketch:

$$\sum_{n=-\infty}^{\infty} x[n-n_0]z^{-n} = z^{-n_0} \sum_{m=-\infty}^{\infty} x[m]z^{-m}.$$

The multiplier z^{-n_0} can add/delete a pole at 0 or at ∞, but it does not move any finite nonzero boundary.

Example

$$\delta[n-1] \xleftrightarrow{\mathcal{Z}} z^{-1}.$$

Scaling, Time Reversal, and Conjugation

Scaling in the z-domain

$$a_0^n x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X\left(\frac{z}{a_0}\right)$$

and the ROC scales by $|a_0|$.

Proof:

$$\sum a_0^n x[n] z^{-n} = \sum x[n] \left(\frac{z}{a_0}\right)^{-n}.$$

Multiplying by $e^{j\omega_0 n}$ rotates the pole-zero pattern; multiplying by r_0^n scales its radii.

Time Reversal

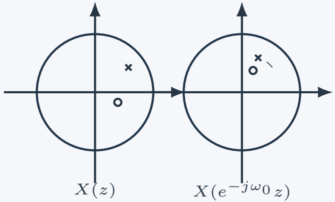
$$x[-n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z^{-1})$$

with inverted ROC.

Proof:

$$\sum x[-n] z^{-n} = \sum_m x[m] (z^{-1})^{-m}.$$

Effect of multiplying by $e^{j\omega_0 n}$



Conjugation

$$x^*[n] \overset{\mathcal{Z}}{\longleftrightarrow} X^*(z^*)$$

with the same ROC.

$$\sum x^*[n] z^{-n} = \left(\sum x[n] (z^*)^{-n}\right)^* = X^*(z^*).$$

For real $x[n]$, $X(z) = X^*(z^*)$, so nonreal poles and zeros occur in conjugate pairs and the ROC is symmetric about the real axis.

Time Expansion

Define the expanded sequence

$$x_{(k)}[n] = \begin{cases} x[n/k], & n \text{ is an integer multiple of } k, \\ 0, & n \text{ is not a multiple of } k. \end{cases}$$

Time expansion property

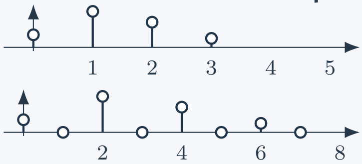
$$x_{(k)}[n] \xleftrightarrow{\mathcal{Z}} X(z^k),$$

with ROC $R^{1/k}$.

Proof from the series:

$$\sum_{n=-\infty}^{\infty} x_{(k)}[n] z^{-n} = \sum_{m=-\infty}^{\infty} x[m] z^{-km} = X(z^k).$$

Insert $k-1$ zeros between samples



ROC Geometry

z is in the new ROC exactly when z^k is in the old ROC, so radii are mapped by $r \mapsto r^k$.

Convolution Property and Proof

Convolution

$$y[n] = x_1[n] * x_2[n] \xleftrightarrow{\mathcal{Z}} Y(z) = X_1(z)X_2(z)$$

with ROC containing at least $R_1 \cap R_2$; cancellation may enlarge it.

Starting from

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k],$$

we obtain

$$\begin{aligned}
Y(z) &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]z^{-n} \\
&= \sum_{k=-\infty}^{\infty} x_1[k]z^{-k} \sum_{m=-\infty}^{\infty} x_2[m]z^{-m} \qquad (m = n-k) \\
&= X_1(z)X_2(z).
\end{aligned}$$

ROC Nuance

Convergence is guaranteed wherever both series converge, so the ROC includes at least $R_1 \cap R_2$. If a pole is cancelled in the product, the final ROC can be larger.

First Difference: Example 10.15

For $h[n] = \delta[n] - \delta[n-1]$,

$$y[n] = x[n] - x[n-1] \overset{\mathcal{Z}}{\longleftrightarrow} (1 - z^{-1})X(z),$$

with ROC at least R except possible deletion of $z = 0$; cancellation may add $z = 1$.

First difference is the discrete-time counterpart of differentiation.

Accumulation

Example 10.16 defines the running sum

$$w[n] = \sum_{k=-\infty}^{n} x[k].$$

Since

$$w[n] = u[n] * x[n],$$

and

$$u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - z^{-1}}, \qquad |z| > 1,$$

we get

$$W(z) = \frac{1}{1 - z^{-1}} X(z).$$

ROC contains at least $R \cap \{|z| > 1\}$.

Intuition: convolution with $u[n]$ is the discrete-time counterpart of integration.

Differentiation in the z-domain

$$n\,x[n] \overset{\mathcal{Z}}{\longleftrightarrow} -z\frac{dX(z)}{dz}$$

with ROC $= R$.

Proof:

$$\frac{dX}{dz} = \sum_n (-n)x[n]z^{-n-1}; \qquad -z\frac{dX}{dz} = \sum_n nx[n]z^{-n}.$$

Example 10.18 Starting from

$$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}},$$

differentiation gives

$$na^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{az^{-1}}{(1-az^{-1})^2}.$$

Linear constant-coefficient difference equation

If

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k],$$

then, using linearity and time shifting,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}.$$

Justification:

$$\mathcal{Z}\{y[n-k]\} = z^{-k}Y(z), \qquad \mathcal{Z}\{x[n-k]\} = z^{-k}X(z).$$

Then factor $Y(z)$ on the left and $X(z)$ on the right.

Book warning

The difference equation determines the **algebraic** system function, but not the ROC. The ROC is extra information: it decides causality, stability, and the impulse response.

Summary of z-Transform Properties

Property	Time-domain signal	z-domain expression	ROC / nuance
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	$\supseteq R_1 \cap R_2$
Time shift	$x[n - n_0]$	$z^{-n_0} X(z)$	same R, except $0, \infty$
Scaling	$a_0^n x[n]$	$X(z/a_0)$	$\|a_0\|R$
Time reversal	$x[-n]$	$X(z^{-1})$	inverted R
Conjugation	$x^*[n]$	$X^*(z^*)$	same R
Time expansion	$x_{(k)}[n]$: insert $k - 1$ zeros	$X(z^k)$	$R^{1/k}$
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	$\supseteq R_1 \cap R_2$
First difference	$x[n] - x[n - 1]$	$(1 - z^{-1})X(z)$	possible 0 deletion, 1 cancellation
Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-z^{-1}} X(z)$	$\supseteq R \cap \{\|z\| > 1\}$
z-domain differentiation	$nx[n]$	$-z\, dX(z)/dz$	same R
Initial value	causal/right-sided from 0	$x[0] = \lim_{z\to\infty} X(z)$	requires $x[n] = 0,\ n < 0$
LCCDE	$\sum a_k y[n - k] = \sum b_k x[n - k]$	$H(z) = \frac{\sum b_k z^{-k}}{\sum a_k z^{-k}}$	equation alone does not set ROC

Analysis of LTI Systems

Transform-Domain View of an LTI System

- For an LTI system with impulse response $h[n]$,

$$Y(z) = H(z)X(z).$$

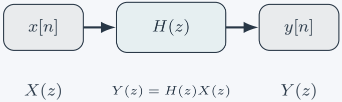
- The z-transform converts convolution into multiplication.
- If the ROC of $H(z)$ includes the unit circle, then

$$H(e^{j\omega})$$

  is the **frequency response**.
- So poles, zeros, and the ROC describe not only the algebra of the transform, but also the behavior of the system.

Main payoff

Just as with Laplace transforms, many system properties become visible directly in the transform domain.



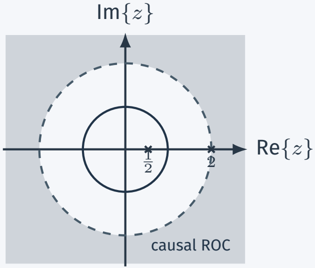
Eigenfunction Reminder

For input $x[n] = z^n$,

$$y[n] = H(z)z^n.$$

Causality Test

A discrete-time LTI system is causal iff the ROC of $H(z)$ is the exterior of a circle and includes ∞.



Example 10.20

$$H(z) = \frac{z^3 - 2z^2 + z}{z^2 + \frac{1}{4}z + \frac{1}{8}}$$

is **not causal** because the numerator degree exceeds the denominator degree when written in powers of z.

Example 10.21

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}, \qquad |z| > 2.$$

The ROC is outside the outermost pole and includes ∞, so the system is causal.

Stability Test

An LTI system is stable iff the ROC of $H(z)$ includes the unit circle.

- **Example 10.22:** the same algebraic $H(z)$ from Example 10.21 is unstable if the ROC is $|z| > 2$.
- **Example 10.23:** for the causal first-order system

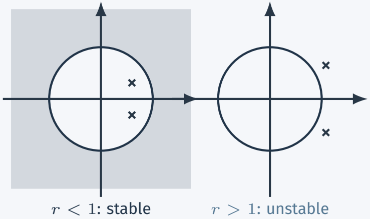
$$H(z) = \frac{1}{1 - az^{-1}},$$

  stability requires $|a| < 1$.
- **Example 10.24:** if

$$H(z) = \frac{1}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}},$$

  then the poles are at $re^{\pm j\theta}$, so a causal system is stable iff $r < 1$.

Complex poles and the unit circle



Example 10.25: Difference Equations Become Algebra

Consider the linear constant-coefficient difference equation

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1].$$

Taking z-transforms and using linearity and time shifting,

$$Y(z) - \frac{1}{2}z^{-1}Y(z) = X(z) + \frac{1}{3}z^{-1}X(z).$$

So

$$Y(z) = X(z)\frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

and therefore

$$H(z) \triangleq \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}.$$

The difference equation fixes the algebraic form of $H(z)$, but not the ROC:

$$|z| > \frac{1}{2} \text{ for the causal interpretation,} \qquad |z| < \frac{1}{2} \text{ for the anticausal one.}$$

Main lesson: a difference equation determines the algebraic form of $H(z)$, but the ROC still determines whether the realization is causal, anticausal, or stable.

Solving a Difference Equation by z-Transform

Use the **causal** system from Example 10.25:

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}, \qquad |z| > \frac{1}{2}.$$

Let the input be the unit step:

$$x[n] = u[n] \xleftrightarrow{\mathcal{Z}} X(z) = \frac{1}{1 - z^{-1}}, \qquad |z| > 1.$$

Then

$$Y(z) = H(z)X(z) = \frac{1 + \frac{1}{3}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - z^{-1}\right)}.$$

Partial fractions give

$$Y(z) = \frac{8/3}{1 - z^{-1}} - \frac{5/3}{1 - \frac{1}{2}z^{-1}}.$$

Hence

$$y[n] = \frac{8}{3}u[n] - \frac{5}{3}\left(\frac{1}{2}\right)^n u[n].$$

This is the discrete-time version of what we did with differential equations in the Laplace domain:

difference equation $\Longrightarrow$ algebra $\Longrightarrow$ inverse transform.